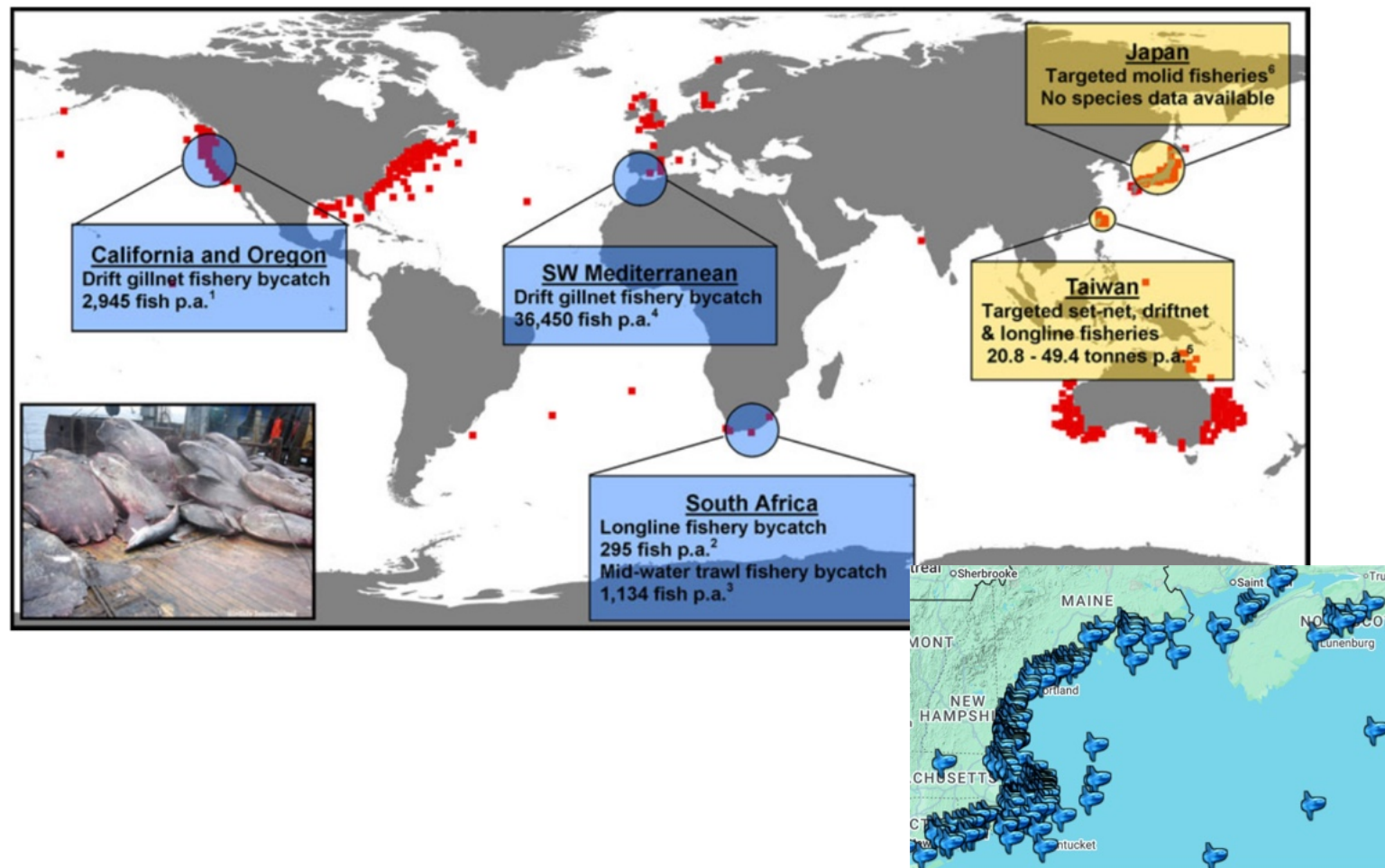


Deepening Research

The biology and ecology of the ocean sunfish *Mola mola*: a review of current knowledge and future research perspectives

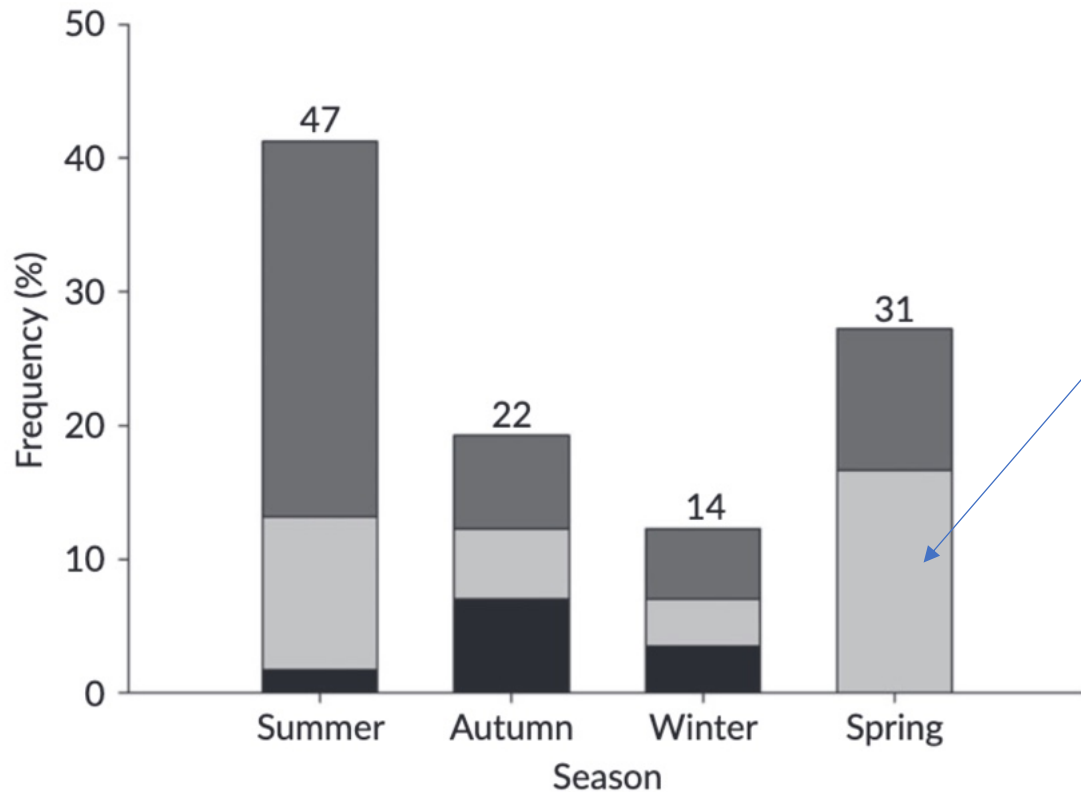
Edward C. Pope • Graeme C. Hays • Tierney M. Thys •
Thomas K. Doyle • David W. Sims • Nuno Queiroz •
Victoria J. Hobson • Lukas Kubicek • Jonathan D. R. Houghton



- World's largest bony fish
- “Little is known” – this came in many studies
- Lack of research probably due to less commercial interest
- Public interest: oceansunfish.org
- Bias: Very little fisheries or survey data, most data is opportunistic, and absence only reflects lack of surveys
- Jellyfish prey is based mainly on surface observations, but molas dive deep
- Overfishing could have removed mola predators and led to more jellyfish
- Common bycatch in S. Africa
- In Spain and California, Molas were more abundant in gillnet fisheries than the target species
- Needs: better understanding of seasonality, diet

The effect of Peruvian small-scale fisheries on sunfishes (Molidae)

Jeffrey C. Mangel^{1,2*}  | Mariela Pajuelo^{1*} | Andrea Pasara-Polack¹ | German Vela³ | Eduardo Segura-Cobeña¹ | Joanna Alfaro-Shigueto^{1,2,4}



- *Mola mola*
- Long thought to feed exclusively on gelatinous prey (Pope et al., 2010), *M. mola* now appear to have a more generalist diet, especially juveniles and sub-adults (Nakamura & Sato, 2014; Sousa et al., 2016)
- *Mola* is threatened due to bycatch in gillnets

Evidence for a Prehistoric *Mola mola* Fishery On the Southern California Coast

JUDITH F. PORCASI,

Zooarchaeology Laboratory, Cotsen Institute of Archaeology, University of California,
Los Angeles, CA 90095-1510

SHERRI L. ANDREWS,

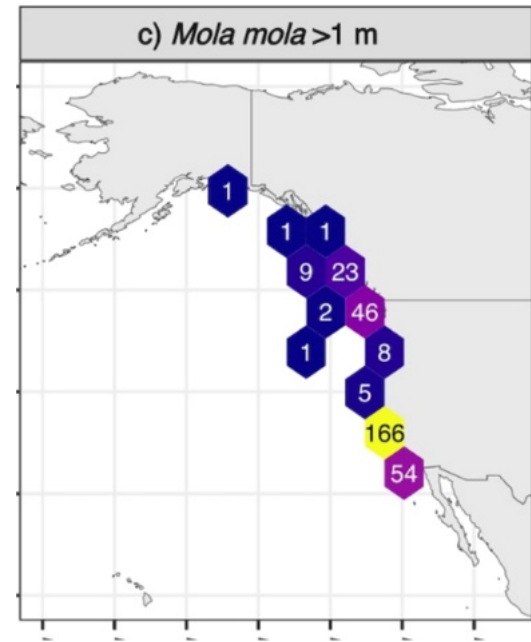
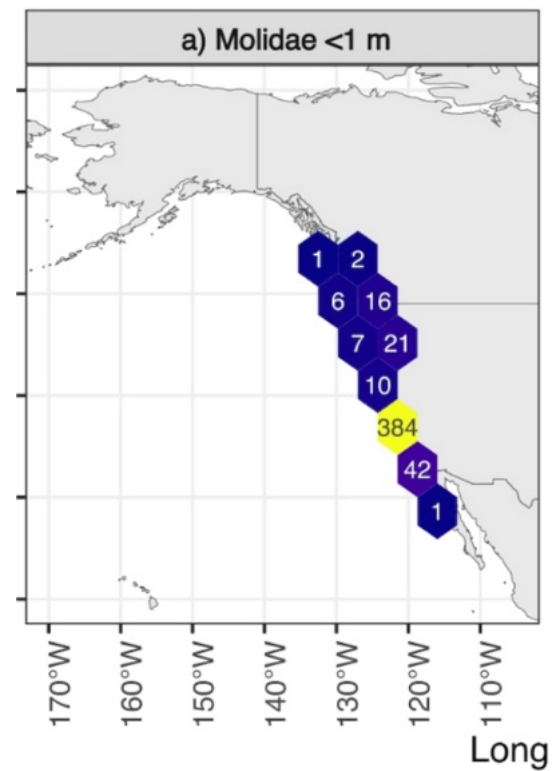
ASM Affiliates, Inc., 543 Encinitas Blvd., Suite 114, Encinitas, CA 92024

- Molas weigh up to 3000 pounds
- They've probably been fished for thousands of years

Hook, line, and social media: crowd-sourced images reveal size and species patterns of ocean sunfishes (Tetraodontiformes, Molidae) from California to Alaska

Tor Mowatt-Larssen¹, Tierney M. Thys², Jackie Hildering³,
Eric J. Caldera⁴, Ellen E. Biesack¹, Jan R. McDowell¹
and Marianne Nyegaard^{5,6*}

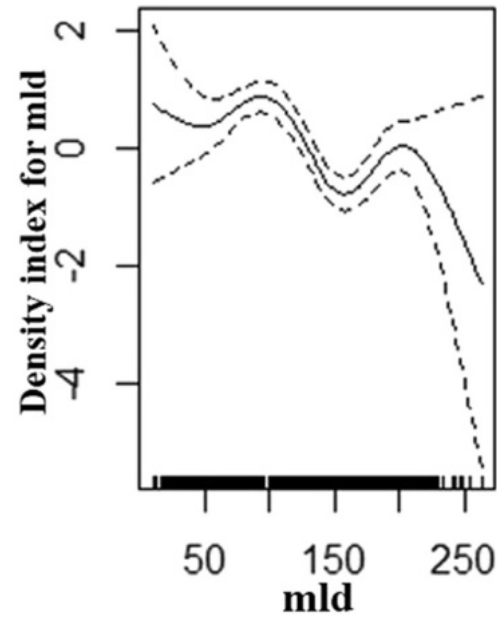
- Utility of community science data
- Different ranges based on size
- South, central California serves as nursery?
- Limitations due to observers



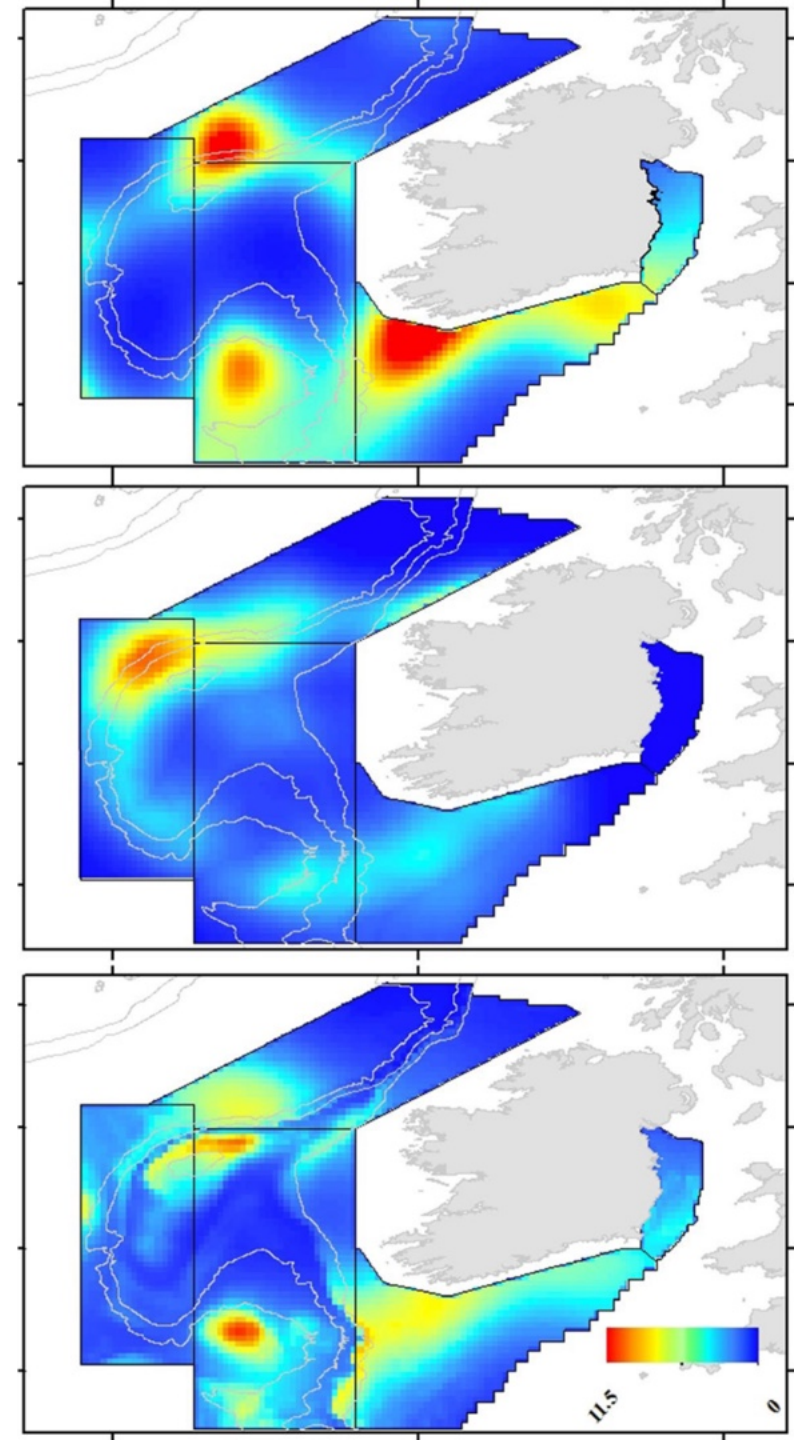
OPEN

New insights into ocean sunfish (*Mola mola*) abundance and seasonal distribution in the northeast Atlantic

15 December 2016
7 April 2017



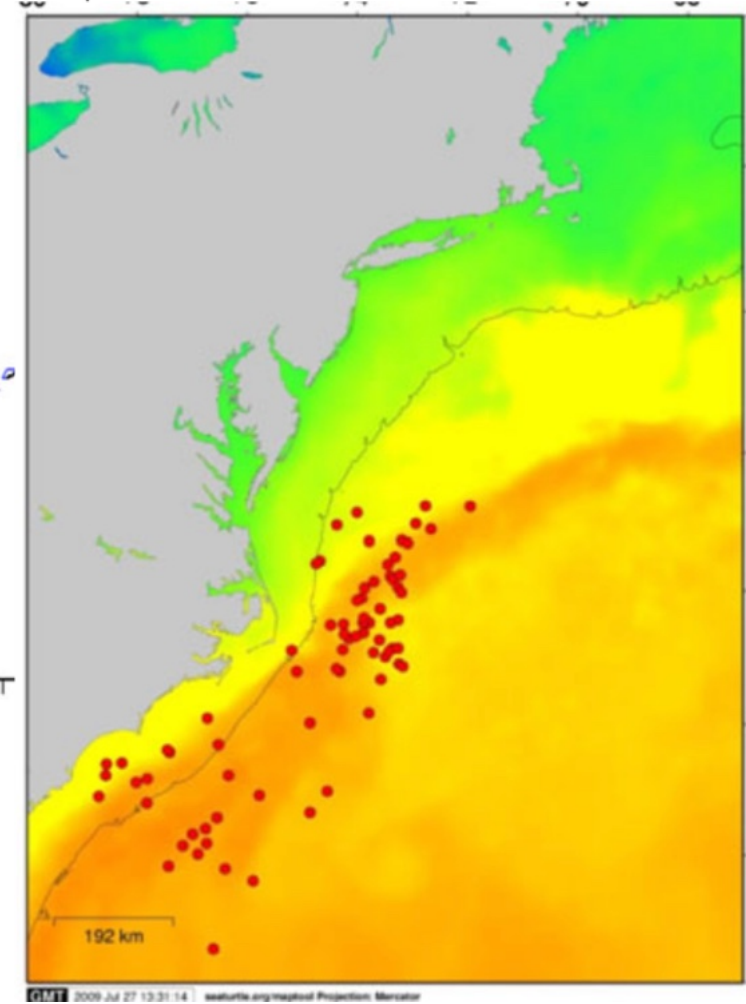
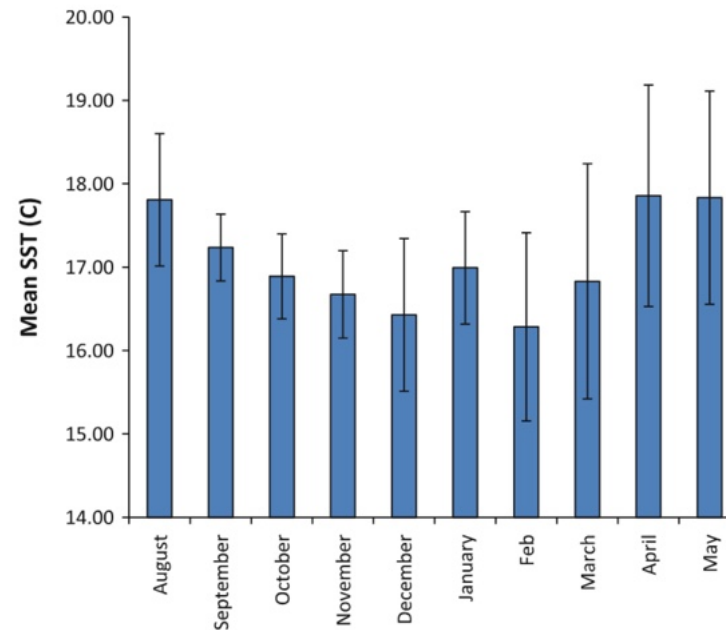
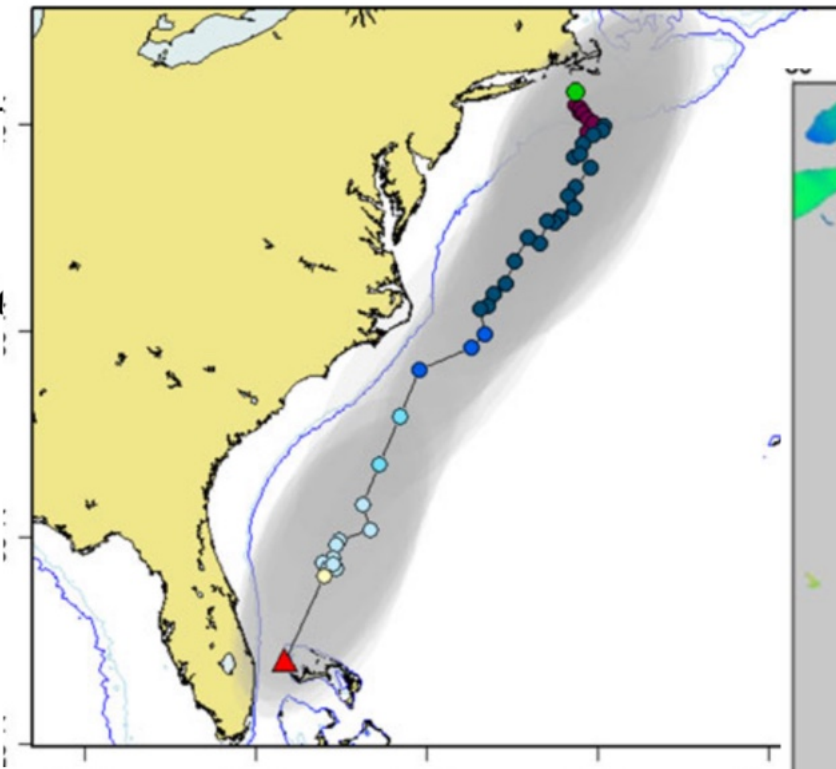
- Estimate a population size of around 8,000
- The model finds **mixed layer depth** as an important covariate
 - Possibly connected to deep dives for foraging
- Estimated daily consumption 2,600 tons of jellyfish!



Horizontal movement of ocean sunfish, *Mola mola*, in the northwest Atlantic

Inga F. Potter · Benjamin Galuardi ·
W. Huntting Howell

- *Mola molas* were tagged in the Gulf of Maine
- Migrations were south as far as the Gulf of Mexico, but seemed to follow constant surface temperature





Environmental influence on the seasonal movements of satellite-tracked ocean sunfish *Mola mola* in the north-east Atlantic

Lara L. Sousa^{1,2,3}, Nuno Queiroz^{1,2}, Gonzalo Mucientes^{2,4}, Nicolas E. Humphries¹ and David W. Sims^{1,3,5*}

- Sea surface temperature and temperature gradients were strong(ish) predictors
- Fish hung out in productive areas (this model included chlorophyll)

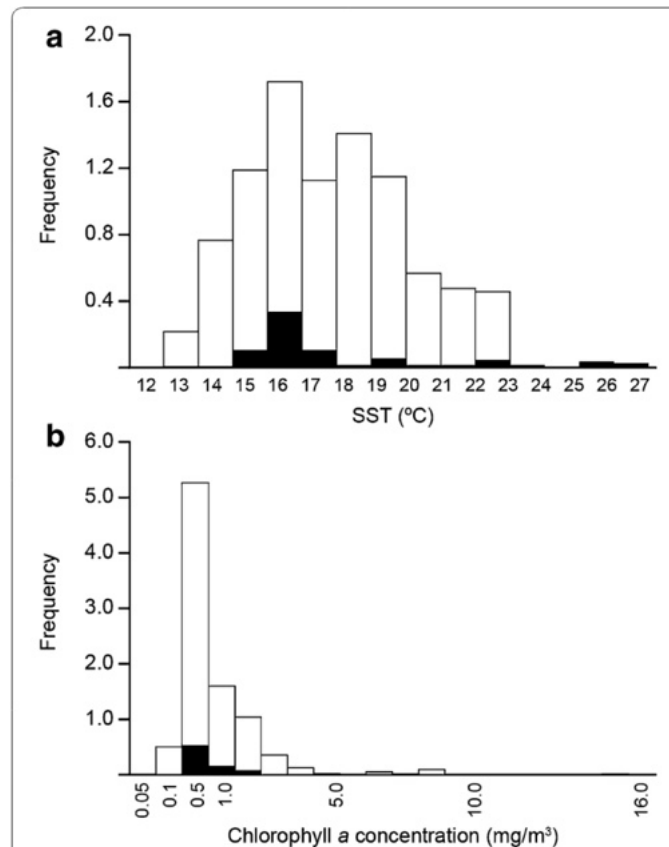
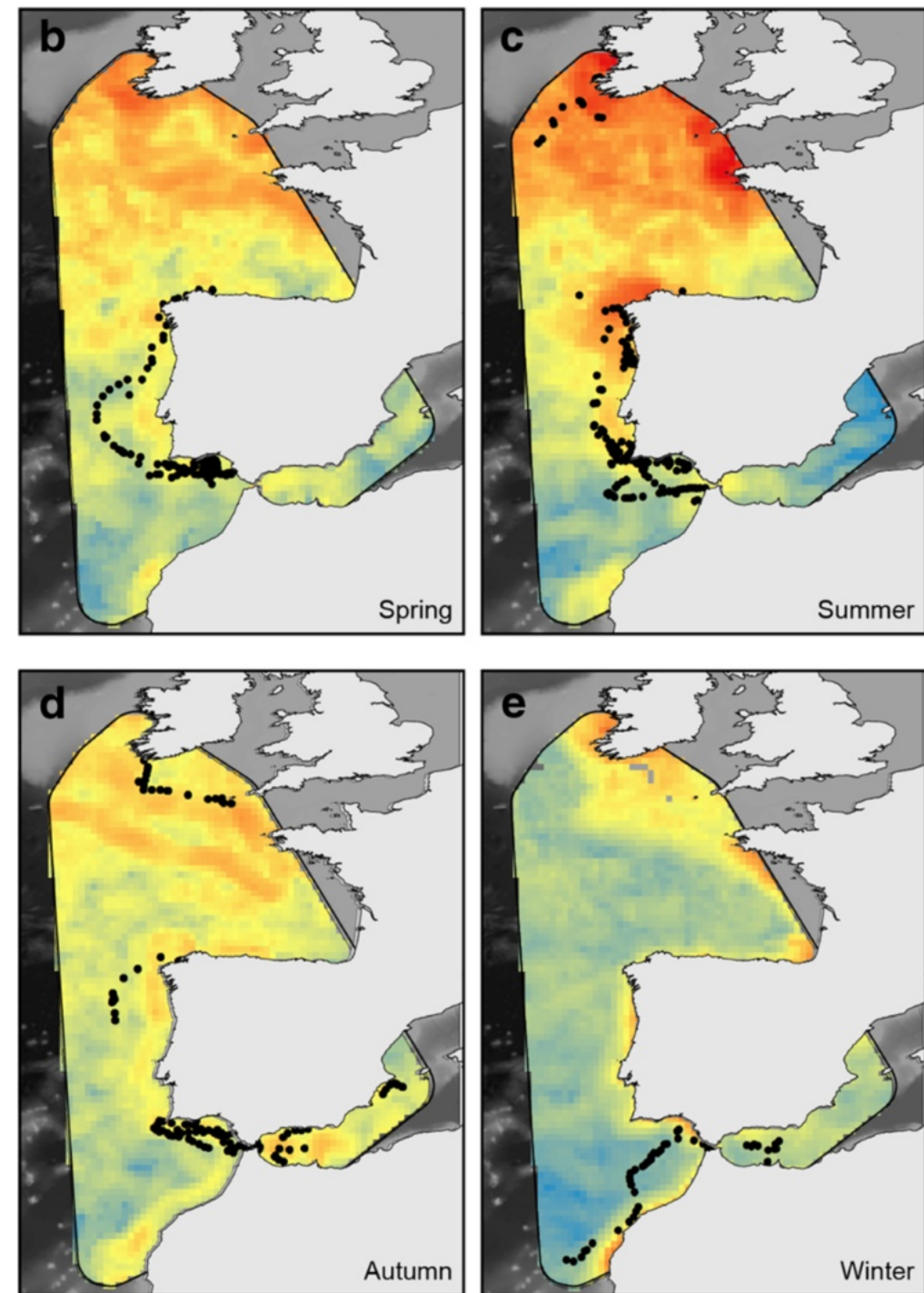
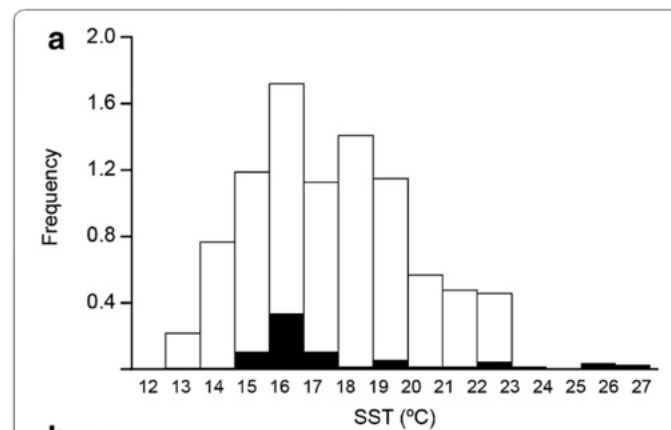
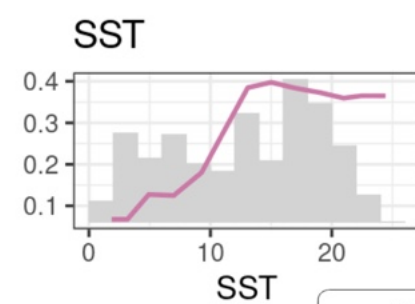
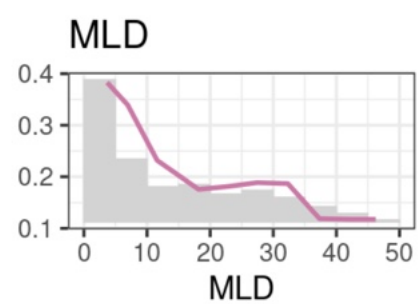
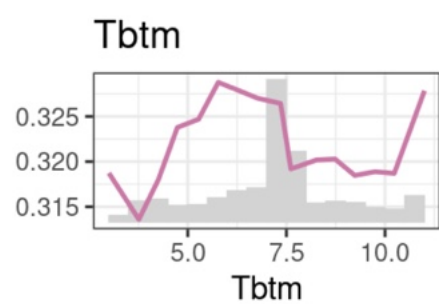
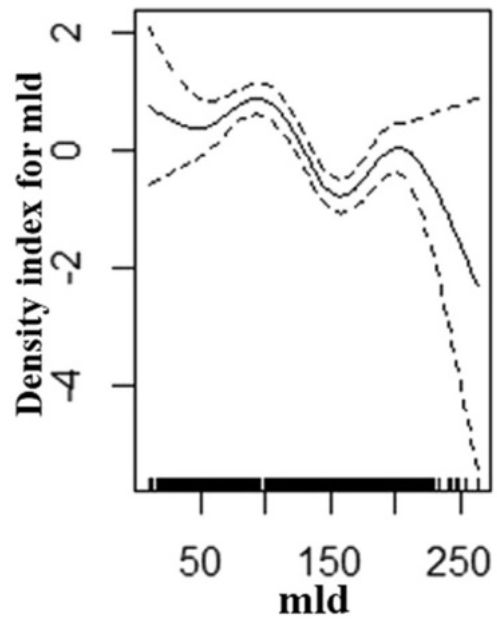
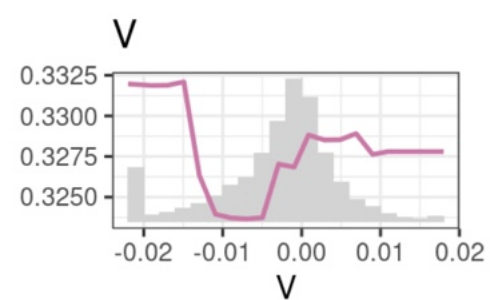
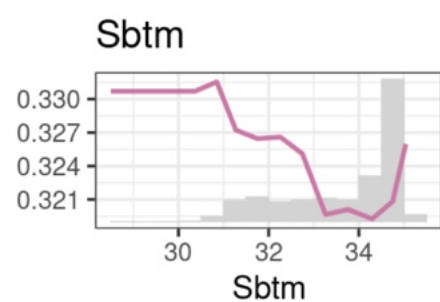
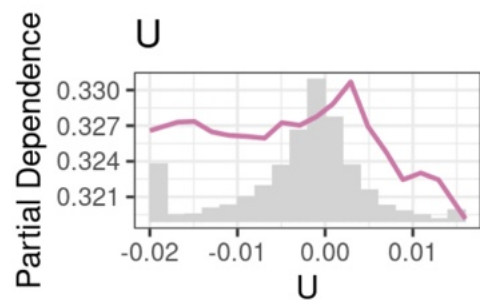
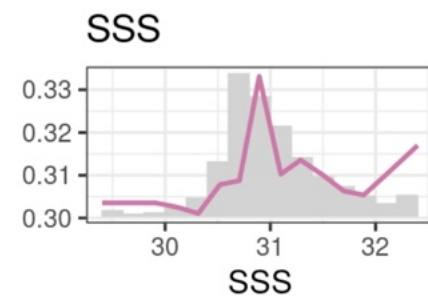
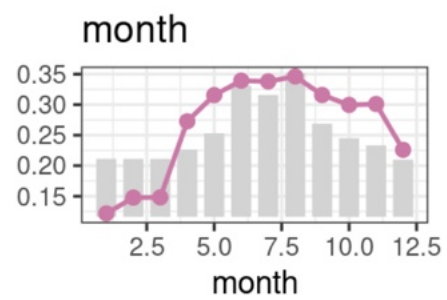
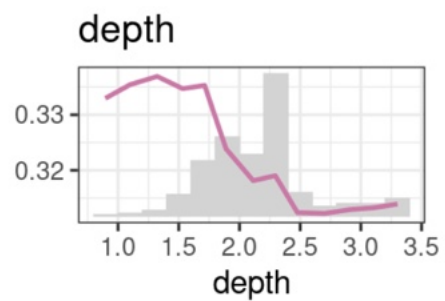


Fig. 4 Encountered SST and chlorophyll *a* histograms for all sunfish tracked locations. *Black* denotes tracks when in the Mediterranean (#14, #17–18) and *white* all the remainders



... so what?



Northeast Ocean Data Portal: <https://www.northeastoceandata.org>

